

Erdős Problem #264

EP-264

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let a_n be a sequence of positive integers such that for every bounded sequence of integers b_n (with $a_n + b_n \neq 0$ and $b_n \neq 0$ for all n) the sum $\sum \frac{1}{a_n + b_n}$ is irrational. Are $a_n = 2^n$ or $a_n = n!$ examples of such a sequence?